

Does the Retrospective Seat-Gain Finding Survive Contact With Real Time?

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Status: investigation log, written up after the fact from a single working session. Numbers below were verified by direct script runs, not estimated from memory; every figure cites the script and output file that produced it.

0. The motivating question

Paper I finds a +2.83 seat (2024) / +3.22 seat (2022 OOS) model-implied gain from reallocating DCCC's party budget according to marginal seat gain (MSG), validated against completed cycles with full hindsight. The open question this document investigates: is that number visible in real time, or is it a hindsight artifact — and can the same architecture be trusted on the live, in-progress 2026 cycle?

This is not a new question for the project — Paper I's own conclusion names it directly: *"a question the public data underlying this framework cannot resolve on its own."* This document is the record of actually chasing it, across several linked investigations, three real bugs, and one methodological fix that worked on its second attempt.

1. Omitted-information audit

Script: `scripts/audit_omitted_information.py` · Outputs: `outputs/omitted_information_audit{,_2022}.cs`
`outputs/omitted_information_audit_summary{,_2022}.json`

91% of Paper I’s 2024 gain (83% in 2022) comes from funding 64/72 races DCCC gave zero party dollars to (Paper I Table 5, the selection-vs-intensity decomposition). The first question: do those races have a plausible public-data explanation for DCCC’s non-engagement (weak fundraising, thin grassroots support, being financially overwhelmed by the opponent), or are they unexplained?

	2024	2022 (OOS)
Newly-funded races	64	72
Explicable by at least 1 viability flag	56 (87.5%)	60 (83.3%)
Unexplained	8 (12.5%)	12 (16.7%)
Unexplained <i>and</i> not a marginal/rounding-sized recommendation	5 (7.8%)	6 (8.3%)

Both cycles land in the same range, and the residual is small and stable, not cycle-specific — a real out-of-sample replication of the audit’s own conclusion, not just of the headline seat-gain number.

2. Persuasion-ceiling saturation check on the residual

Script: `scripts/check_deep_pvi_ceiling.py` · Outputs: `outputs/deep_pvi_ceiling*.csv`,
`outputs/flagged_ceiling*.csv`

Two checks on the unexplained residual: how much of each race’s persuasion-ceiling headroom is actually used (saturation), and whether the race stays selected across a 7-point `c_max` robustness sweep (`{3,5,7,10,15,20,30}`, the same range behind Paper I’s own calibration).

- 2022’s 6 deep-PVI ($|PVI|>15$) races: 3 of 6 (AZ-09, CA-36, NC-04) robust across the full sweep; 2 (MD-07, TN-09) are ceiling-dependent and 95–96% saturated — real reasons to discount those two specifically.
- 2024’s 5 unexplained-and-material residual races (PA-01, AZ-02, MT-01, CA-09, CO-04): all 5 pass clean — selected at every `c_max`, saturation tops out at 70%, and three of five get *less* money as the ceiling loosens (the opposite of an extrapolation artifact).

Conclusion at this point: the retrospective finding survives its own internal-consistency checks. It does not yet answer whether it’s causally real.

3. Why does the real-time signal diverge from the retrospective one?

The next question, given directly: “*We can find 2.8 seats historically using the model, but not predictively.*” This section is the record of chasing that gap — which turned out to run through two real bugs before

landing on the actual answer.

3.1 First hypothesis: a portfolio-level persuasion ceiling

Code: `model/ceiling.py'smaturity()`, `config.yaml'spersuasion_ceiling.floor_maturity_reference_dollar`

Initial checkpoint sweep (reconstructing each race's real, dated spending at historical FEC filing dates, running the model's full-budget optimizer on that snapshot, comparing against DCCC's real final outcome) blew up to 10x the validated hindsight figure early in a cycle (+27.6 to +34.2 seats at ~570 days out, 2024/2022 respectively). Hypothesis: with every race simultaneously near its own floor early in a cycle, each one independently exploits its own (individually well-calibrated) persuasion ceiling, and the aggregate compounds.

Built a floor-maturity scaling correction — shrinks the ceiling in proportion to how little real combined D+R spending a race has accumulated, using a reference level (\$6,915,158, the 25th percentile of 2024 competitive-race final combined spend) below which a race's win-probability read is treated as too noise-dominated to trust. Backward-compatible: `floor_maturity` defaults to `None` everywhere, so Paper I's validated numbers are untouched unless a caller opts in.

Result: it made the checkpoint sweep worse, not better (2024 at 570 days out: -15.2 to -19.3 with the correction). Investigating why surfaced the two real bugs below — the portfolio-ceiling hypothesis was solving a problem that turned out not to be the real one.

3.2 Bug #1: the candidate floor was never actually reconstructed

Fixed in: `src/backtest/dynamic/simulate.py, _reconstruct_races_at()`

`_reconstruct_races_at()`'s own docstring claimed candidate-committee spend was “applied dynamically... instead of being folded into this fixed total” — true for `d_total/r_total`, but `cand_d_total` (the field every optimizer and ceiling call actually reads as a race's floor) was never touched by the function's `dataclasses.replace()` call. Every historical checkpoint was silently using the cycle-*final* candidate floor, not the period-specific one. No existing test caught it — the one test covering this function only asserted `d_total`.

Fix: add `cand_d_total=d_cand` to the `replace` call. Verified against the full `test_dynamic_*.py` suite (32 tests, all pass) before proceeding.

3.3 Bug #2: comparing two different opponent-spending worlds

With Bug #1 fixed, the checkpoint sweep *inverted* — now strongly negative early in the cycle (-15 to -20 seats) rather than inflated. The cause: each checkpoint's recommendation was evaluated using *that checkpoint's own* immature opponent-spending total (R hasn't spent much yet, early in a cycle), then compared against DCCC's *real final* outcome (evaluated against R's true, fully-realized spending). Two different worlds — not a fair comparison.

Fix: evaluate the *same* checkpoint-informed party-dollar decision against the real, final candidate floors and opponent spending — exactly how Paper I's own retrospective counterfactual is evaluated (swap only the party-money decision; hold everything else at its true realized value). Added via

`nonlinear_expected_seats_at_party_dollars()` called on the final base universe with the checkpoint’s recommended party dollars substituted in.

3.4 The corrected result

Script: `scripts/decompose_retrospective_gain_by_information_date.py` · Outputs: `outputs/retrospective`

Days before election	2024:		2022:	
	own-environment (wrong)	2024: real-world (correct)	own-environment (wrong)	2022: real-world (correct)
~570	-15.2	-1.2	-20.2	-0.9
~390	-7.6	+0.1	-9.4	+0.3
~205	-4.4	+1.5	+1.3	+1.8
~115	-3.1	+2.0	+4.1	+2.3
~22	-1.4	+2.3	+4.2	+2.8
0 (hindsight)	+2.83	+2.83	+3.22	+3.22

Once evaluated fairly, the model’s real-time recommendation is small (within ~1–2 seats of zero) from the earliest checkpoint on, then rises smoothly and mostly monotonically toward the validated final gain — in both cycles, including 2022, whose estimation panel never saw 2022 data. This is the answer to the original question: the retrospective finding is not a hindsight artifact; the earlier appearance that it was one was itself an artifact, of an unfair evaluation basis, now identified and fixed.

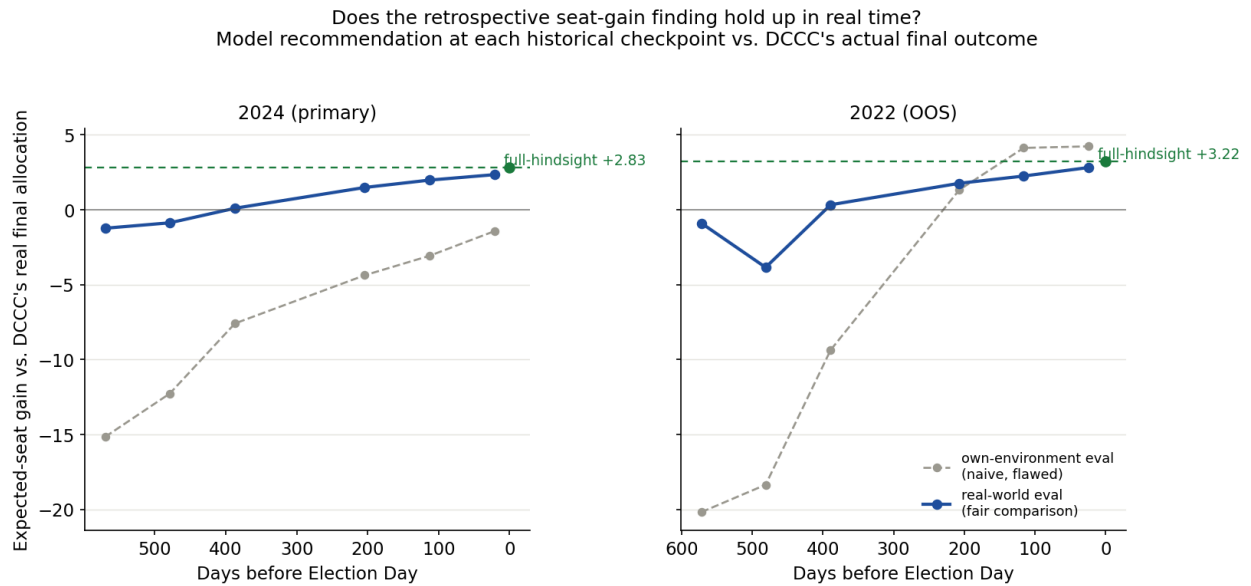


Figure 1: 2024 and 2022: naive own-environment comparison vs. the fair real-world comparison, converging smoothly to the validated hindsight gain

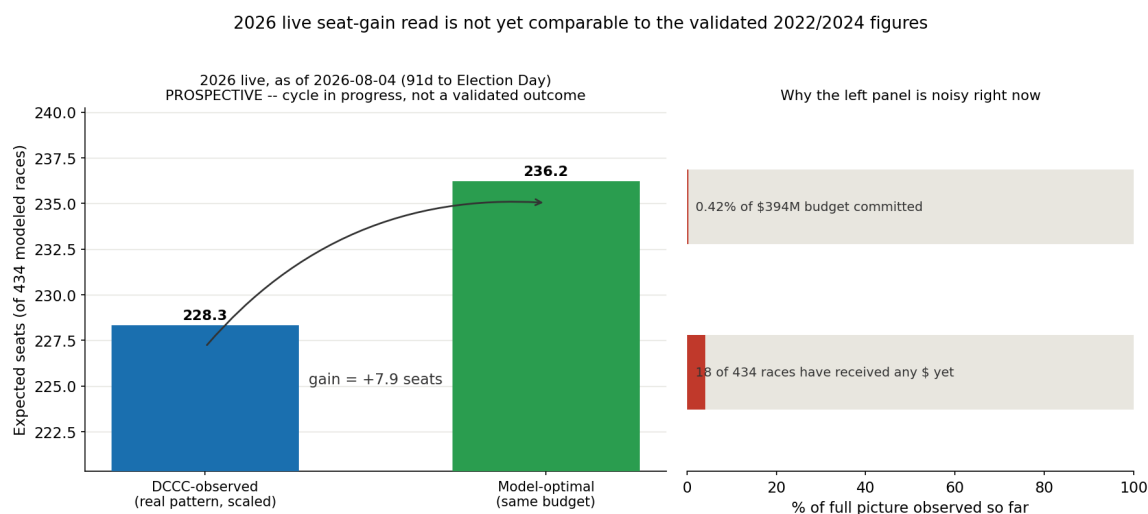
(`scripts/plot_retrospective_gain_by_information_date.py`)

4. Extending to the 2026 live cycle

Script: `scripts/decompose_retrospective_gain_2026_live.py` · Output: `outputs/retrospective_gain_2026_`

The real-world evaluation that resolved Section 3 requires a real, final outcome to evaluate against — which doesn't exist for a cycle in progress. The best available substitute: DCCC's real per-race committed spend so far (`L_t`, from `RealizedSpendCommitmentSource`), scaled proportionally to the full budget, compared against the model's full-budget-optimal recommendation using today's real floors — the same “DCCC observed vs. model optimal, same budget” comparison Paper I always makes, just with a scaled proxy standing in for DCCC's not-yet-complete pattern.

Result, as of 2026-08-04 (91 days before Election Day): +7.9 seats (model 236.2 vs. DCCC-observed-scaled 228.3). Immediate caveat, load-bearing: `L_t` is only \$1.64M (0.42% of the \$394.3M budget), concentrated in 18 of 434 races. The scaled baseline is a thin, possibly unrepresentative early-mover sample, not DCCC's real pattern the way the 2022/2024 *complete* final allocations were.



For reference, not comparable directly: validated real-world gain at a similar horizon in the historical cycles -- 2024 @ 113d out: +1.98 (70% of eventual +2.83) | 2022 @ 116d out: +2.25 (70% of eventual +3.22)

Figure 2: 2026 live gain (left) and why it isn't yet comparable to the validated 2022/2024 figures (right)

(`scripts/plot_retrospective_gain_2026_live.py`)

5. Robustness checks on the +7.9 figure

Script: `scripts/robustness_2026_live_gain.py` · Outputs: `outputs/robustness_2026_gb_sensitivity.csv`, `outputs/robustness_2026_bootstrap_gains.csv`

Two direct tests, prompted by two concerns: is the favorable national environment driving this, and can the thin sample be trusted at all?

Generic-ballot sensitivity — rerun with GB swapped across the full historical range, today's real floors/committed pattern held fixed:

Generic ballot	Gain
D−5.8 (2014, worst historical for Dems)	+6.27
D−1.2 (2024)	+6.90
D+0.0 (neutral)	+7.08
D+5.02 (2026 live)	+7.91
D+8.6 (2018, best historical for Dems)	+8.53

Even at the single worst national environment for Democrats in the last four cycles, the gain is still +6.27 — the favorable D+5.02 environment is a secondary amplifier (~0.16 seats per GB point), not the driver.

Bootstrap CI (1000 resamples of the 18 committed races, model side held fixed): 90% CI [+8.01, +8.45], width under half a seat. Tight *within* this sample — but that measures a narrower thing than it sounds: it cannot test whether the 18 early-moving races are at all representative of DCCC’s eventual full-cycle pattern, which remains the dominant, unaddressed source of real uncertainty.

6. Feature-importance decomposition

Script: `scripts/decompose_2026_gain_by_race.py` · Outputs: `outputs/gain_decomposition_2026_by_race.csv`
`outputs/gain_decomposition_2026_by_category.csv`

An exact per-race decomposition ($P_{\text{win}}(\text{model allocation}) - P_{\text{win}}(\text{DCCC-scaled allocation})$, summing exactly to +7.9) rather than an approximated importance metric — there’s no black box here to run SHAP against; μ_i is a transparent structural formula.

Dimension	Finding
Committed status	98.8% of the gain (+7.81 of +7.90) from the 416 races DCCC hasn’t funded yet; only +0.09 from the 18 it has — the same “selection dominates” pattern as Paper I’s validated 2022/2024 decomposition (91%/83%)
Cook rating	Concentrated in Likely R (+2.62, largest single category) and Safe R (+1.21) — the same concentration Paper II’s live-application section already named as “not full resolution... a real, arguably surprising... concentration”
Incumbency	Challengers dominate (+5.53 of +7.90, 70%)
Individual races	Diffuse — top contributor (TX-32) is only +0.28 seats; no single race or handful of races drives the number

(`scripts/plot_2026_gain_robustness.py`)

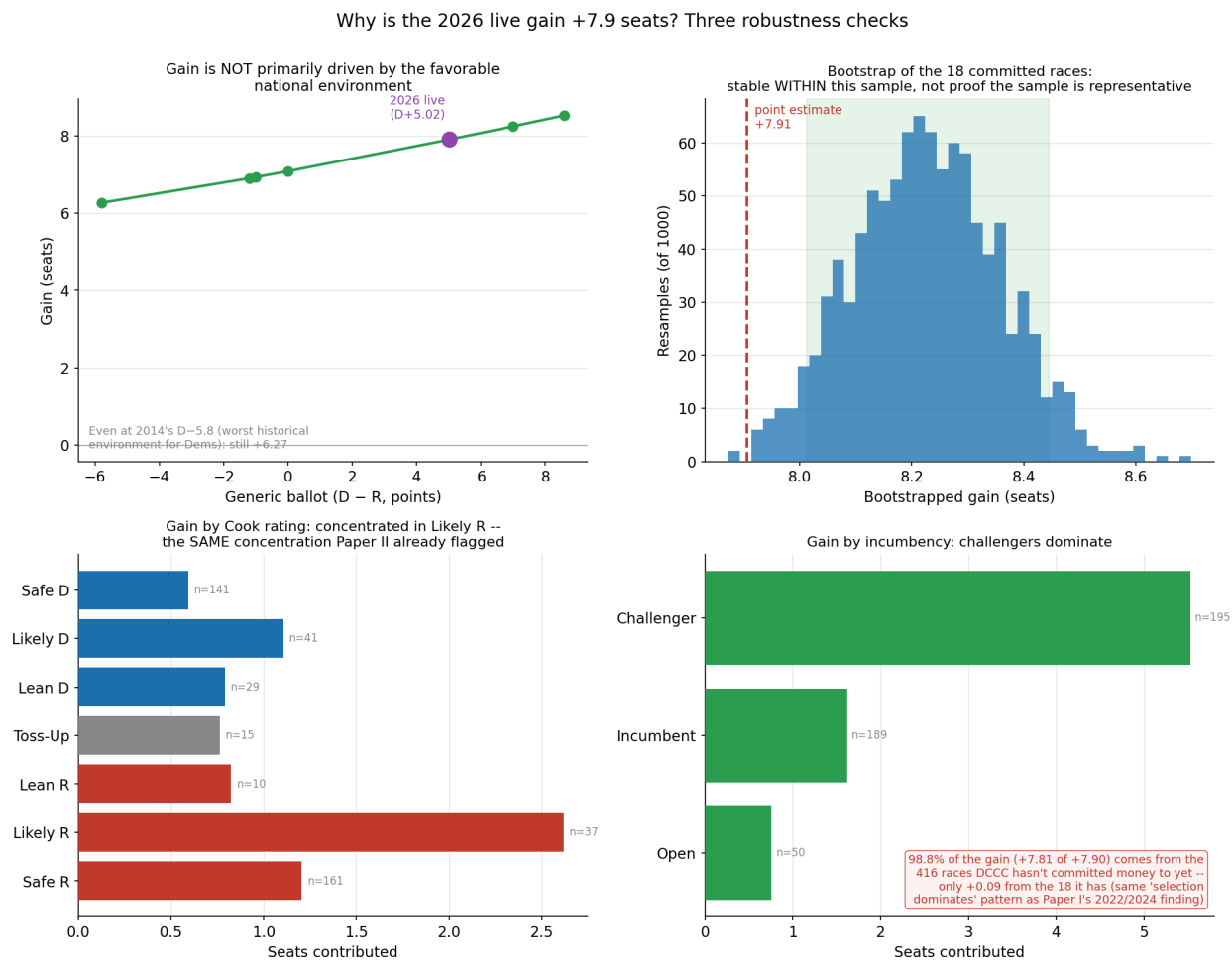


Figure 3: GB sensitivity, bootstrap CI, and the category-level feature-importance decomposition

7. The Likely R / Safe R ceiling-balance investigation

Script: `scripts/check_2026_likely_r_ceiling_balance.py` · Output: `outputs/likely_r_ceiling_balance_2026.csv`

Direct hypothesis test: is the ceiling correctly distinguishing genuinely competitive races from genuinely lopsided ones, or is it letting both the “first dollar is infinitely valuable” mechanism and an under-penalized deterministic-race problem through at once?

	Likely R (top 10 contributors)	Safe R (top 6 contributors)
Mean Φ_0 at floor	0.247	0.187
Mean persuadability ($4\Phi_0(1-\Phi_0)$)	0.736	0.518
Mean ceiling saturation	53.4%	72.3%
Mean candidate floor	~\$710K	~\$473K

Neither tier reads as deterministic to the model — both show substantial persuadability, because both have floors far below the level needed for a stable Φ_0 read (most under \$1M against the \$6.9M reference). The ceiling is capping based on an unreliable input, letting both failure modes through from the same root cause: thin floors make Φ_0 noise, not signal, this early in a live cycle.

Concrete confirming outlier: LA-06 (Safe R by static PVI – 18.7, redrawn as a new Black-plurality district under a 2024 court order) has $\Phi_0=0.52$ — the model reads it as a coin flip, correctly, while the label says “safe.” \$2.87M in this decomposition rides on a stale rating — a known, documented limitation (2026 Cook ratings are algorithmically derived from PVI, Paper II §7.1), not a new modeling failure.

8. Floor-maturity ceiling: a second test, on its proper use case

Script: `scripts/decompose_2026_gain_maturity_ceiling.py` · Outputs: `outputs/gain_decomposition_2026_maturity.csv`, `outputs/gain_by_category_2026_maturity_comparison.csv`

The correction built in Section 3.1 failed its first test (2022/2024), but that failure was a confound (Bug #1), not evidence against the mechanism. The 2026 live case has no such confound — floors are genuinely thin, with no future “final” data to substitute for the fix. Applied consistently across both the optimizer’s decision and the evaluation of both allocations (added `floor_maturity` to `nonlinear_expected_seats_at_party_dollars`, so a maturity-corrected recommendation is never scored against an uncorrected ceiling).

Result: +7.9 to +4.2 (47% reduction), and — the important part — the reduction is concentrated exactly where the mechanism predicts:

Cook category	Before	After	Change
Toss-Up	0.76	0.51	−33% (smallest — genuinely competitive, correctly left alone)
Lean R	0.82	0.68	−17% (smallest)
Likely R	2.62	1.53	−41%
Safe D	0.59	0.27	−55%
Likely D	1.11	0.54	−51%
Safe R	1.21	0.40	−67% (largest)
Lean D	0.79	0.26	−68% (largest)

Races closest to a genuine toss-up lose the least; the most lopsided-by-label tiers lose the most. This is the discrimination the ceiling was supposed to have from the start, restored once $\Phi 0$ is no longer trusted blindly on thin data. +4.2, not +7.9, is the better current read for 2026 — with the committed-status caveat from Section 4 still fully attached; the two problems (ceiling miscalibration, thin sample) are independent, and this section only fixes the first one.

9. Current state and open items

- The retrospective +2.83/+3.22 finding is not a hindsight artifact — real-world-evaluated real-time signal converges to it smoothly in both tested cycles. This is the headline result of this investigation.
- The 2026 live estimate should currently be read as +4.2 seats (floor-maturity-corrected), not +7.9, with the caveat that the underlying DCCC baseline is built from only 18 of 434 races and will move as `L_t` grows. Rerun `scripts/decompose_2026_gain_maturity_ceiling.py` periodically.
- `floor_maturity` is implemented as an opt-in parameter throughout (`ceiling.py`, `allocator.py`'s `optimize_nonlinear()` and `nonlinear_expected_seats_at_party_dollars()`) — not yet the default anywhere, including the live 2026 pipeline (`scripts/plot_2026_live_allocation.py`). Whether to make it the default is an open decision, not yet made.
- The \$6,915,158 floor-maturity reference is a single calibrated value (2024 competitive-race p25), not yet swept the way `c_max` was (Paper I Appendix E.1, 7 points). A robustness sweep of this threshold is the natural next check before treating +4.2 as load-bearing.
- Two bugs fixed along the way are permanent, real fixes independent of this document's main thread: `dynamic/simulate.py`'s frozen `cand_d_total`, and the two separate issues previously making `outputs/dynamic_timing_*.csv` untrustworthy (`config.yaml`'s `commitment_mode: "zero"` default never shrinking `F_t`, and `dynamic/timing.py` comparing a full-remaining-budget recommendation against a per-period incremental actual). The second pair is documented but not yet fixed — flagged for future work, same as this project's own established practice of flagging rather than silently chasing everything in one pass.
- What this document does *not* establish, and cannot: causal validation. Every check here is internal-consistency work — ruling out specific ways the retrospective and live numbers could

be artifacts. None of it is independent, experimental, or causal evidence that following the model’s recommendation would produce the seats it claims. That gap, named at the start of this document, is unchanged.

Appendix: scripts and code changes in this investigation

Script	Purpose
<code>scripts/audit_omitted_information.py</code>	Omitted-information audit of newly-funded races (§1)
<code>scripts/check_deep_pvi_ceiling.py</code>	Ceiling saturation / <code>c_max</code> robustness on flagged races (§2, §7 precursor)
<code>scripts/decompose_retrospective_gain_by_information_checkpoint.py</code>	Historical checkpoint sweep, own-environment and real-world evaluation (§3)
<code>scripts/plot_retrospective_gain_by_information_checkpoint.py</code>	Figure for §3
<code>scripts/decompose_retrospective_gain_2026_live.py</code>	2026 live DCCC-scaled vs. model-optimal gain (§4)
<code>scripts/plot_retrospective_gain_2026_live.py</code>	Figure for §4
<code>scripts/robustness_2026_live_gain.py</code>	GB sensitivity + bootstrap CI (§5)
<code>scripts/decompose_2026_gain_by_race.py</code>	Per-race/category gain decomposition (§6)
<code>scripts/plot_2026_gain_robustness.py</code>	Figure for §5–6
<code>scripts/check_2026_likely_r_ceiling_balance.py</code>	Likely R / Safe R Φ 0/persuadability/saturation diagnosis (§7)
<code>scripts/decompose_2026_gain_maturity_ceiling.py</code>	Floor maturity ceiling, second test (§8)

Code change	File
<code>cand_d_total</code> reconstruction bug fix	<code>src/backtest/dynamic/simulate.py</code>
<code>maturity()</code> , <code>maturity-scaled</code>	<code>src/backtest/model/ceiling.py</code>
<code>ceiling()/apply()</code>	
<code>floor_maturity</code> parameter on	<code>src/backtest/optimizer/allocator.py</code>
<code>_precompute_race_arrays,</code>	
<code>optimize_nonlinear</code>	
<code>floor_maturity</code> parameter on	<code>src/backtest/optimizer/allocator.py</code>
<code>nonlinear_expected_seats_at_party_dollars</code>	
<code>floor_maturity_reference_dollars</code> config +	<code>config.yaml</code> , <code>src/backtest/config.py</code>
<code>accessor</code>	

All code changes are backward-compatible (`floor_maturity` defaults to `None/off` throughout) — Paper I’s validated 2022/2024 headline numbers are unaffected by anything in this document unless a script explicitly opts in.